

Jesús C. Blázquez Martínez

Computer Science & Mathematics Student

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EDUCATION

Double Major in Computer Science & Mathematics

Expected 2028

Universidad Autónoma de Madrid (UAM)

- Cumulative GPA: 9.14/10. Distinction in Computer Architecture (2026) and Data Structures (2024); Academic Excellence Scholarship (2024).
- Coursework: Operating Systems, Computer Architecture, Algorithm Design, Networks, Numerical Analysis, Linear Algebra.

Erasmus Exchange Student in Computer Science

Oct 2026 – Aug 2027

Technical University of Munich (TUM)

- Planned: Computer Systems, High Performance Computing, Cybersecurity, Artificial Intelligence.

Professional Degree in Music (Oboe)

2013 – 2023

Conservatorio Tomás de Torrejón y Velasco

TECHNICAL SKILLS

Languages: C++ (C++17/20, STL), C, Python, Bash, Lua, Java, LaTeX
Systems: Linux/Unix admin, epoll, POSIX threads, memory management, networking, Docker
Tooling: GDB, Valgrind, AddressSanitizer, Make/CMake, Git, Vim/Neovim
Mathematics: Numerical Analysis, Linear Algebra, Probability, Vector Analysis
Spoken: Spanish (native), English (professional working proficiency)

ENGINEERING & SYSTEMS PROJECTS

POSTHaste — Custom C++ HTTP Server | C++17 | Networking | epoll

 jesuscbm/posthaste

- Built a zero-dependency HTTP/1.1 server from scratch in C++17 (Reactor pattern), benchmarked at 123k req/s (219 MB/s static throughput).
- Implemented a non-blocking, edge-triggered epoll event loop with a thread pool for concurrent request handling.

Memory Allocator (malloc clone) | C | Memory Management

 jesuscbm/memory-allocator

- Implemented a from-scratch dynamic memory allocator in C to study heap layout, fragmentation and free-list management.

FreeBSD libm Contributions | C | Numerical Analysis | Open Source

 freebsd/freebsd-src

- Authored and merged floating-point routines into the upstream FreeBSD standard math library, targeting C23 conformance.
- Ensured bit-exact accuracy and robust handling of numerical edge cases (NaN, Infinity).

Multi-threaded Sorting Visualizer | C | SDL3 | Concurrency

 jesuscbm/sorting_visualizer

- Developed a real-time C/SDL3 rendering engine visualizing algorithms such as QuickSort, using POSIX threads to keep the UI responsive.

RESEARCH EXPERIENCE

Summer Research Fellow (SURF)

Summer 2026

Institute for Computer Science, Artificial Intelligence and Technology (INSAIT)

- Selected as one of 15 students globally from an applicant pool of 2,800+ for a competitive research fellowship (Sofia, Bulgaria).
- Fuzzed the multi-threaded C++ i2pd codebase and diagnosed memory corruption, race conditions and leaks with GDB, Valgrind and AddressSanitizer.
- Discovered a vulnerability in i2pd (CVE ID [CENSORED], reserved); technical details [CENSORED] pending coordinated disclosure.
- First author of a security research paper on the I2P anonymity network. Title and details [CENSORED] pending double-blind peer review (in preparation).

EXTRACURRICULARS & INTERESTS

HomeLab & Systems Administration

Ongoing

Personal Infrastructure

- Maintains a self-hosted Linux homelab (Docker, VPN, reverse proxy); daily-drives a customized Arch Linux workstation.